Install tensorflow-gpu 2.5.0 and CUDA 11.2 with CuDNN 8.1.0, Drivers 460, and RTX 3090 for artificial intelligence.

Adarsh Singh Dikhit · Follow
3 min read · Jan 10, 2023

The RTX3090 was used to test this lesson on Ubuntu 20.04 LTS. The installation manual that is referenced is available at: https://medium.com/analytics-vidhya/install-cuda-11-2-cudnn-8-1-0-and-python-3-9-on-rtx3090-for-deep-learning-fcf96c95f7a1.

1. **Check version**

Build from source | TensorFlow
Build a TensorFlow pip package from source and install it on Ubuntu Linux and macOS. While the instructions might work…
www.tensorflow.org

2. Remove previous installation

```
sudo apt-get purge nvidia*
sudo apt remove nvidia-*
sudo rm /etc/apt/sources.list.d/cuda*
sudo apt-get autoremove && sudo apt-get autoclean
sudo rm -rf /usr/local/cuda*
```

3. Verify your GPu is cuda enable check

```
lspci | grep -i nvidia
```

if gcc compiler is not there , install

```
sudo apt install build-essential
sudo apt -y install gcc-8 g++-8 gcc-9 g++-9
sudo update-alternatives --install /usr/bin/gcc gcc /usr/bin/gcc-8 8
sudo update-alternatives --install /usr/bin/g++ g++ /usr/bin/g++-8 8
sudo update-alternatives --install /usr/bin/gcc gcc /usr/bin/gcc-9 9
sudo update-alternatives --install /usr/bin/g++ g++ /usr/bin/g++-9 9
sudo update-alternatives --config gcc
```

4. Configuration

Ubuntu setup by running the following commands.

```
sudo apt-get update
sudo apt-get upgrade -ysudo apt-get install -y build-essential cmake unzip pkg-c
sudo apt-get install -y libxmu-dev libxi-dev libglu1-mesa libglu1-mesa-dev
sudo apt-get install -y libjpeg-dev libpng-dev libtiff-dev
sudo apt-get install -y libavcodec-dev libavformat-dev libswscale-dev libv4l-dev
sudo apt-get install -y libxvidcore-dev libx264-dev
sudo apt-get install -y libgtk-3-dev
sudo apt-get install -y libopenblas-dev libatlas-base-dev liblapack-dev gfortran
sudo apt-get install -y libhdf5-serial-dev graphviz
sudo apt-get install -y python3-dev python3-tk python-imaging-tk
sudo apt-get install -y linux-image-generic linux-image-extra-virtual
sudo apt-get install -y linux-source linux-headers-generic
```

5. Install Nvidia-Drivers

Step 5.1 Add Graphic Drivers PPA

```
sudo add-apt-repository ppa:graphics-drivers/ppa
sudo apt-get update
```

Step 5.2 Search available drivers

```
ubuntu-drivers devices
```

Step 5.3 : Install the driver with the best version

```
sudo apt-get install nvidia-driver-460
```

Step 5.4: Reboot your computer after installation.

```
Sudo reboot
```

Step 6: Install Cuda 11.2

CUDA Toolkit 11.2 Downloads
Select Target Platform Click on the green buttons that describe your target platform. Only supported platforms will be…
developer.nvidia.com

Installation Phase:

1. Select operation system as Linux

2. Select Architecture x86_64

3. Select Distribution as Ubuntu

4. Select Version as 20.04

5. Select Installer type as deb(local)

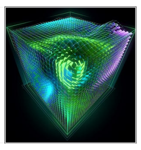
```
wget https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2004/x86_64/
sudo mv cuda-ubuntu2004.pin /etc/apt/preferences.d/cuda-repository-pin-600
wget https://developer.download.nvidia.com/compute/cuda/11.2.0/local_installers/
sudo dpkg -i cuda-repo-ubuntu2004-11-2-local_11.2.0-460.27.04-1_amd64.deb
sudo apt-key add /var/cuda-repo-ubuntu2004-11-2-local/7fa2af80.pub
sudo apt-get update
sudo apt-get -y install cuda
```

Step 7: Setup your paths

```
echo 'export PATH=/usr/local/cuda/bin:$PATH' >> ~/.bashrc
echo 'export LD_LIBRARY_PATH=/usr/local/cuda/lib64:$LD_LIBRARY_PATH' >> ~/.bashr
source ~/.bashrc
sudo ldconfig
```

Step 8: Install Cudnn v8.1

CUDA Deep Neural Network
The NVIDIA CUDA® Deep Neural Network library (cuDNN) is a GPU-accelerated library of primitives for deep neural…
developer.nvidia.com



Download cuDNN v8.1.0 (released on January 26th, 2021), for CUDA 11.0,11.1, and 11.2 from the link above. Then, unzip the embedded folder.

```
tar -zvxf cudnn-11.2-linux-x64-v8.1.0.77.tgz
```

copy the following files into the cuda toolkit directory.

```
sudo cp -P cuda/include/cudnn.h /usr/local/cuda/include
sudo cp -P cuda/lib64/libcudnn* /usr/local/cuda/lib64/
```

Finally, reset the read and write permissions of the cudnn.h file.

```
sudo chmod a+r /usr/local/cuda/include/cudnn.h
```

Step 9: Verify the installation

```
nvidia-smi  #check cuda
nvcc -V  #check cudnn
```

Step 10: Install Tensorflow-Gpu

```
pip3 install tensorflow-gpu==2.5.0
```

Enjoy !!!

You can reach out to me via LinkedIn if any help is required.

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AI/ML Engineer skilled in Apache Kafka, Python, LLMs, Generative AI, Docker, ETL, and AWS. Passionate about building scalable AI solutions and data pipelines.

Responses (1)

srinidhi ramesh
Jan 10, 2023

Nice. Very descriptive and helpful!

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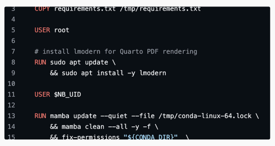

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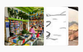

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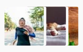

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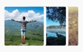

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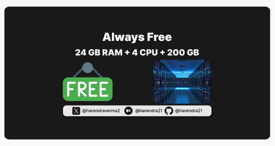

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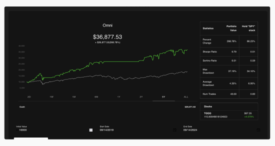

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